

Target and stitched multiple-shooting estimate

actual common support: 19.009–23.959 s

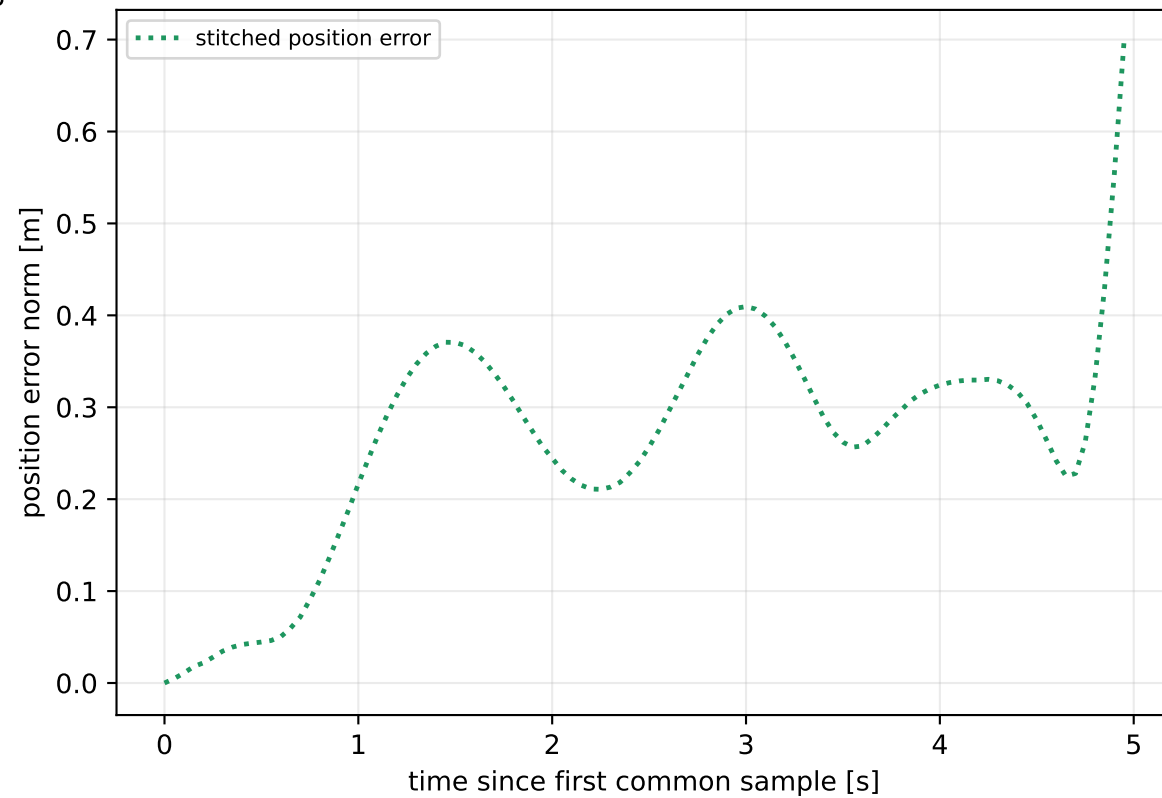
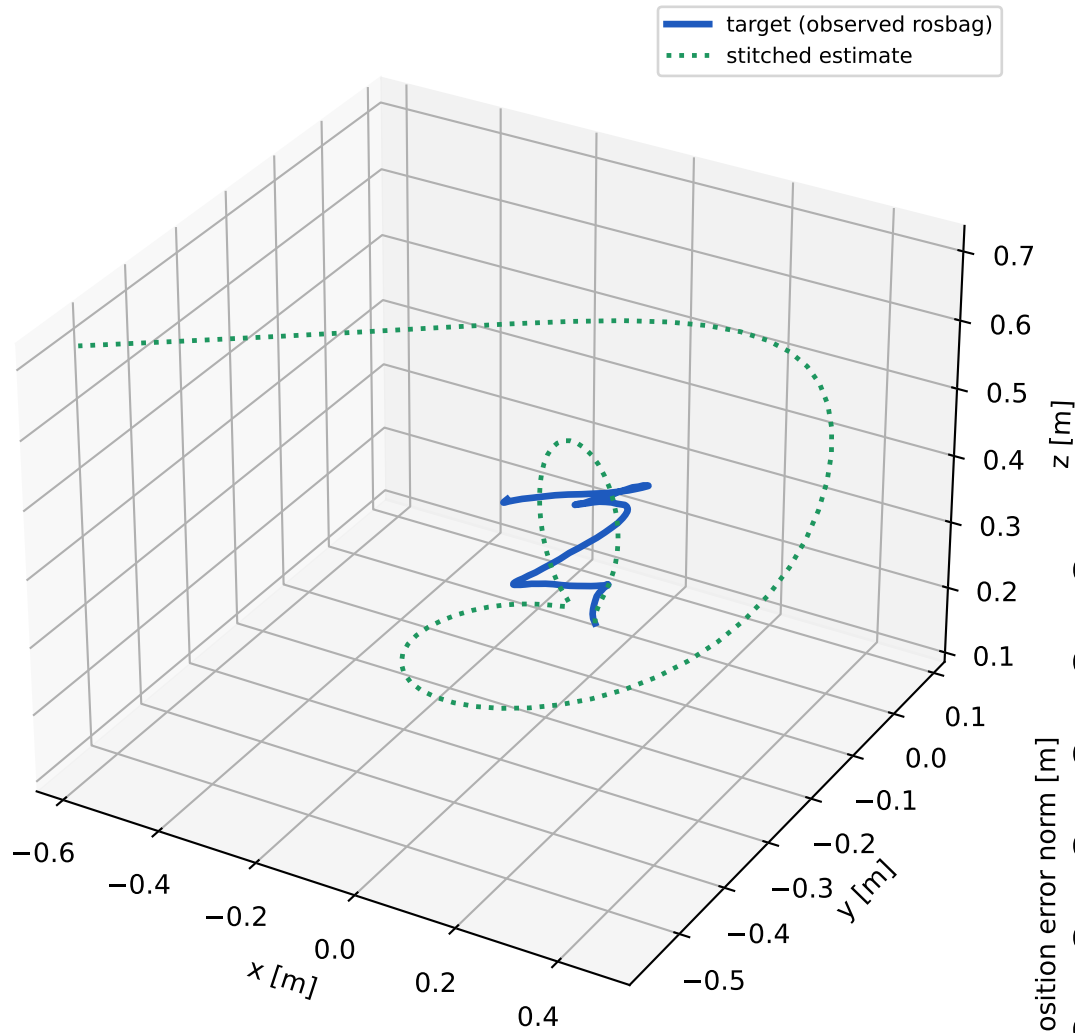
selected command delay: 0.02 s

metric	stitched
position RMSE [m]	0.2911
orientation RMSE [deg]	20.99
velocity RMSE [m/s]	0.84703
angular velocity RMSE [rad/s]	0.63221
specific force RMSE [m/s ²]	1.7998

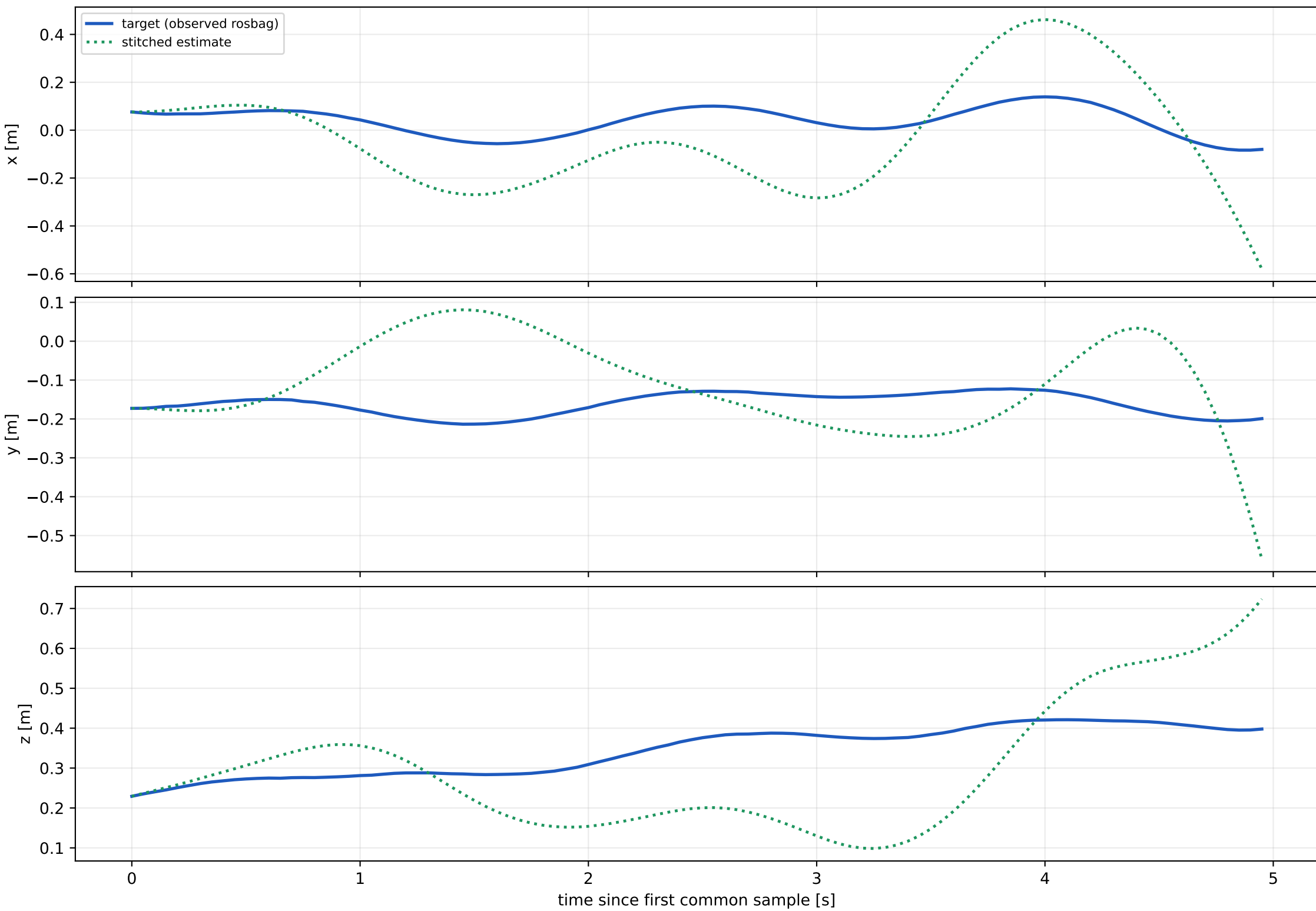
continuity max / tolerance:
0.0002279 / 0.001 (converged)

Velocity and IMU metrics are diagnostic only;
the fit uses sensor position and orientation.

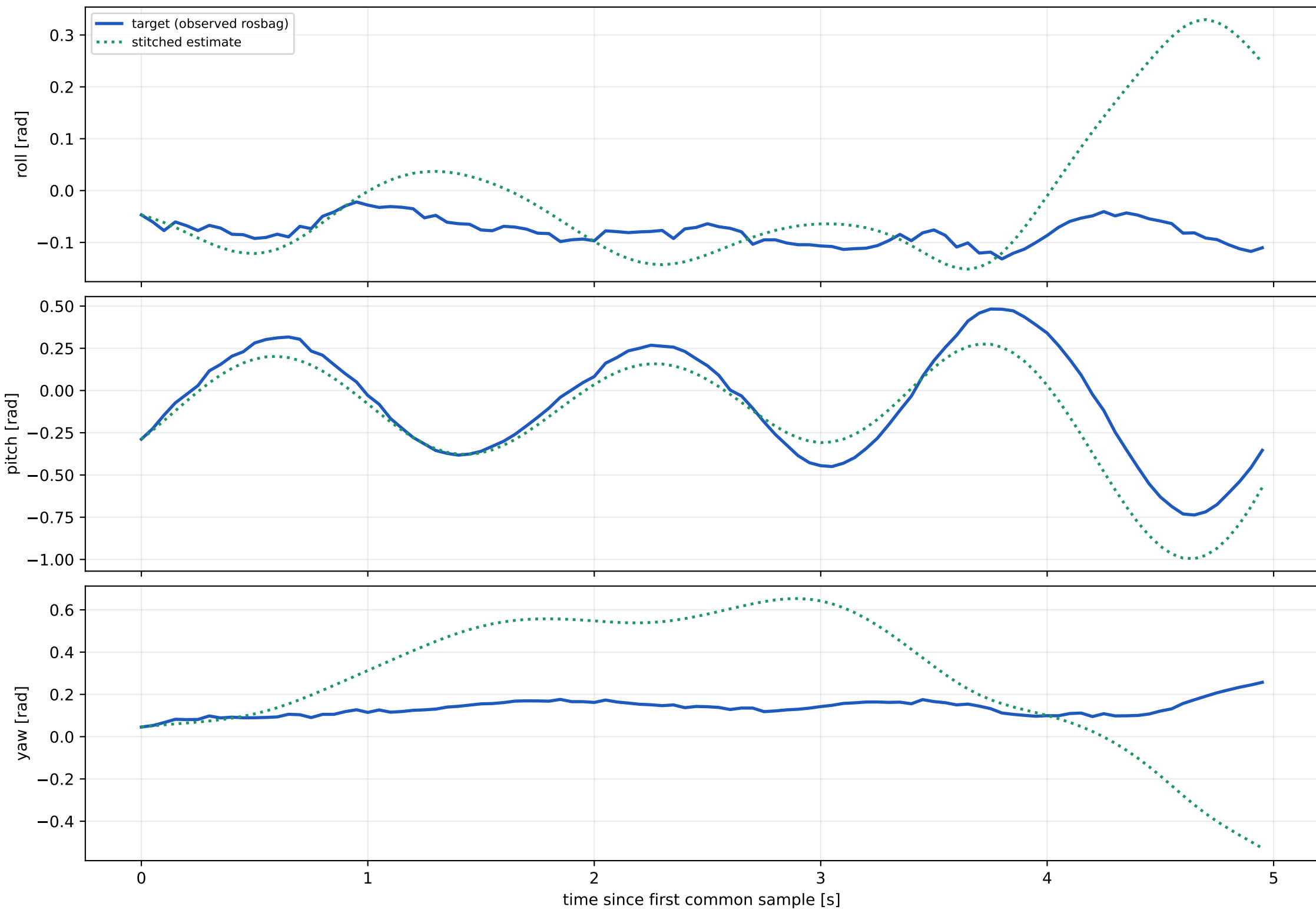
Stitched trajectory



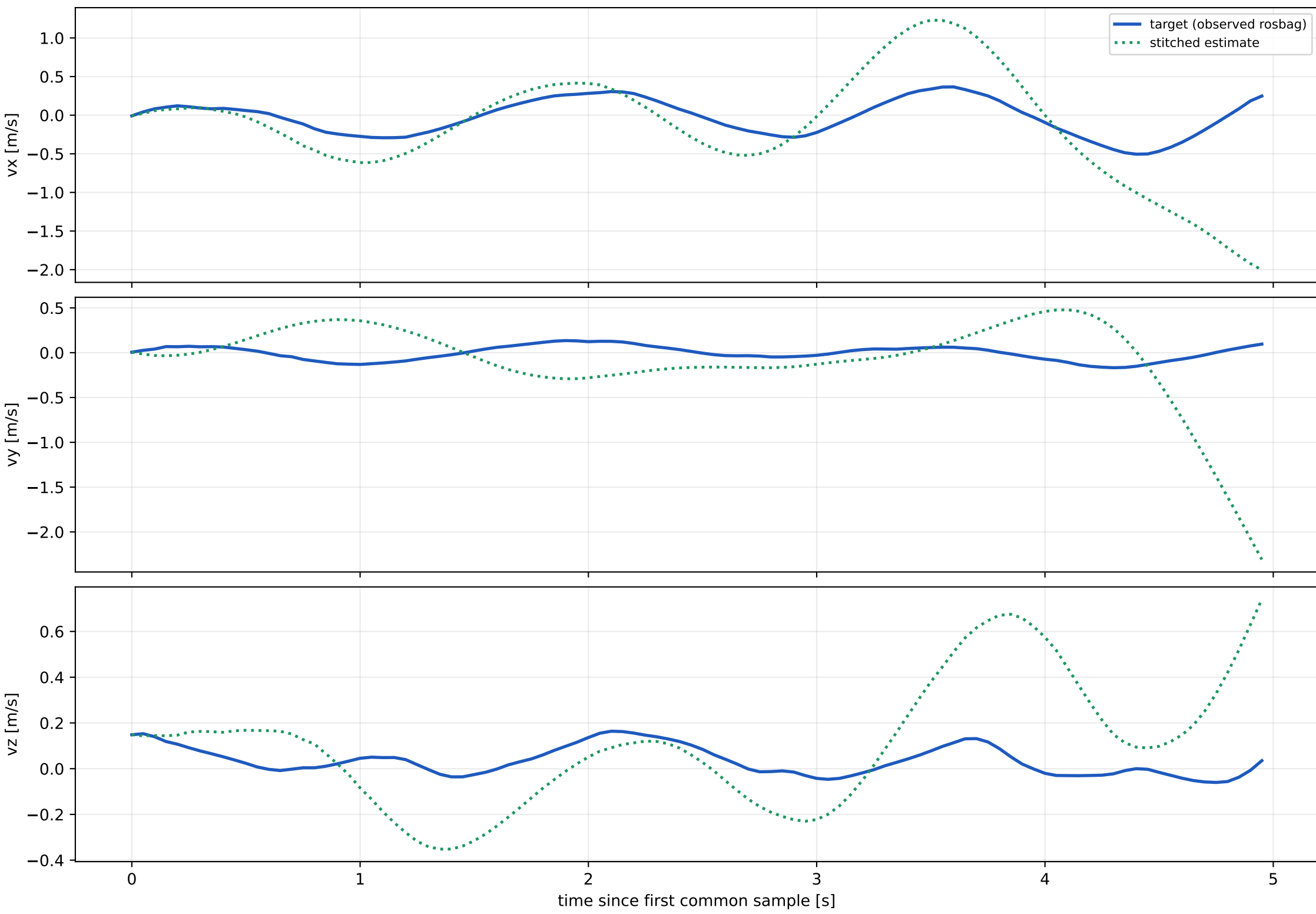
Sensor position



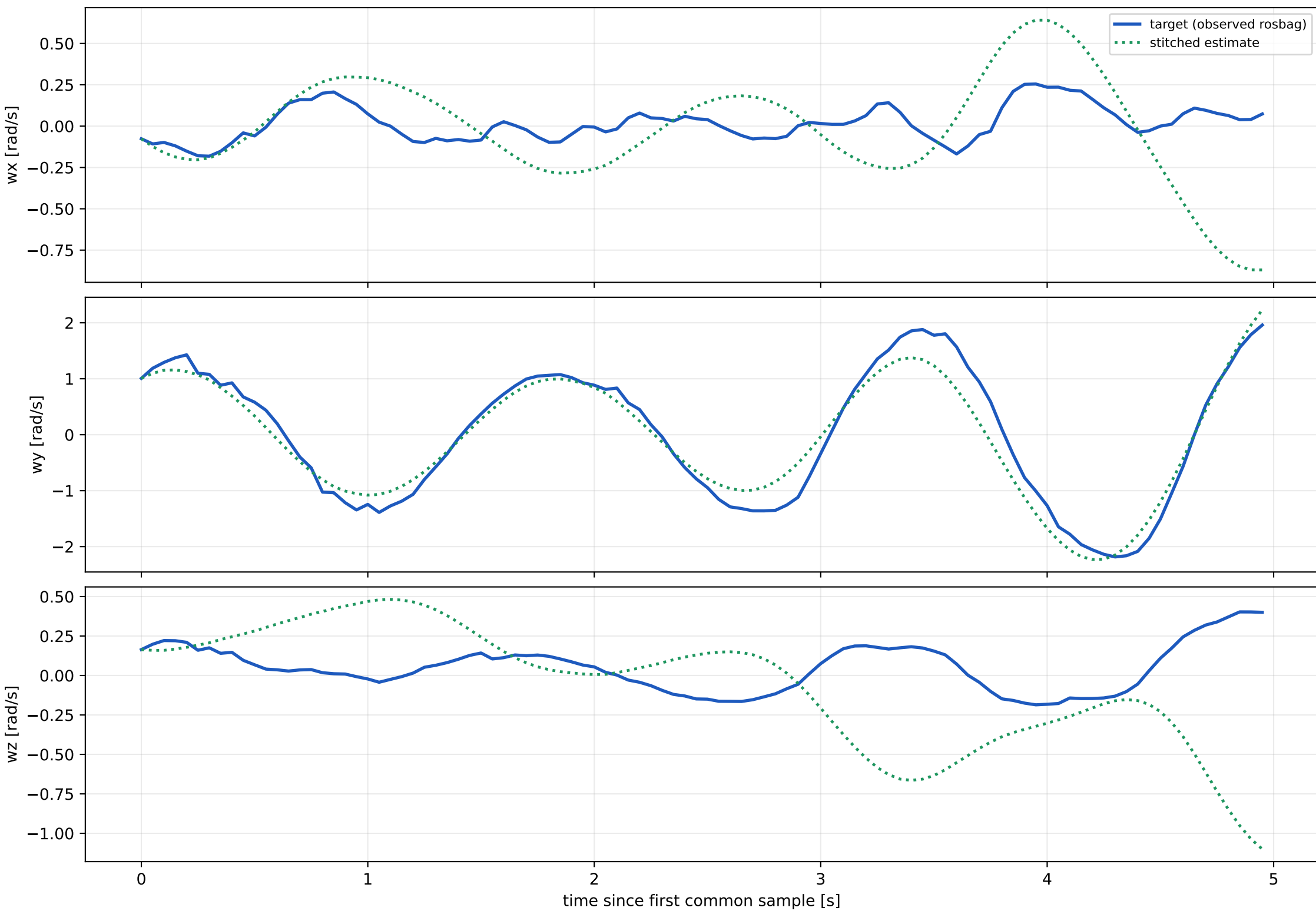
Sensor orientation (display only; fit uses SO(3) log)



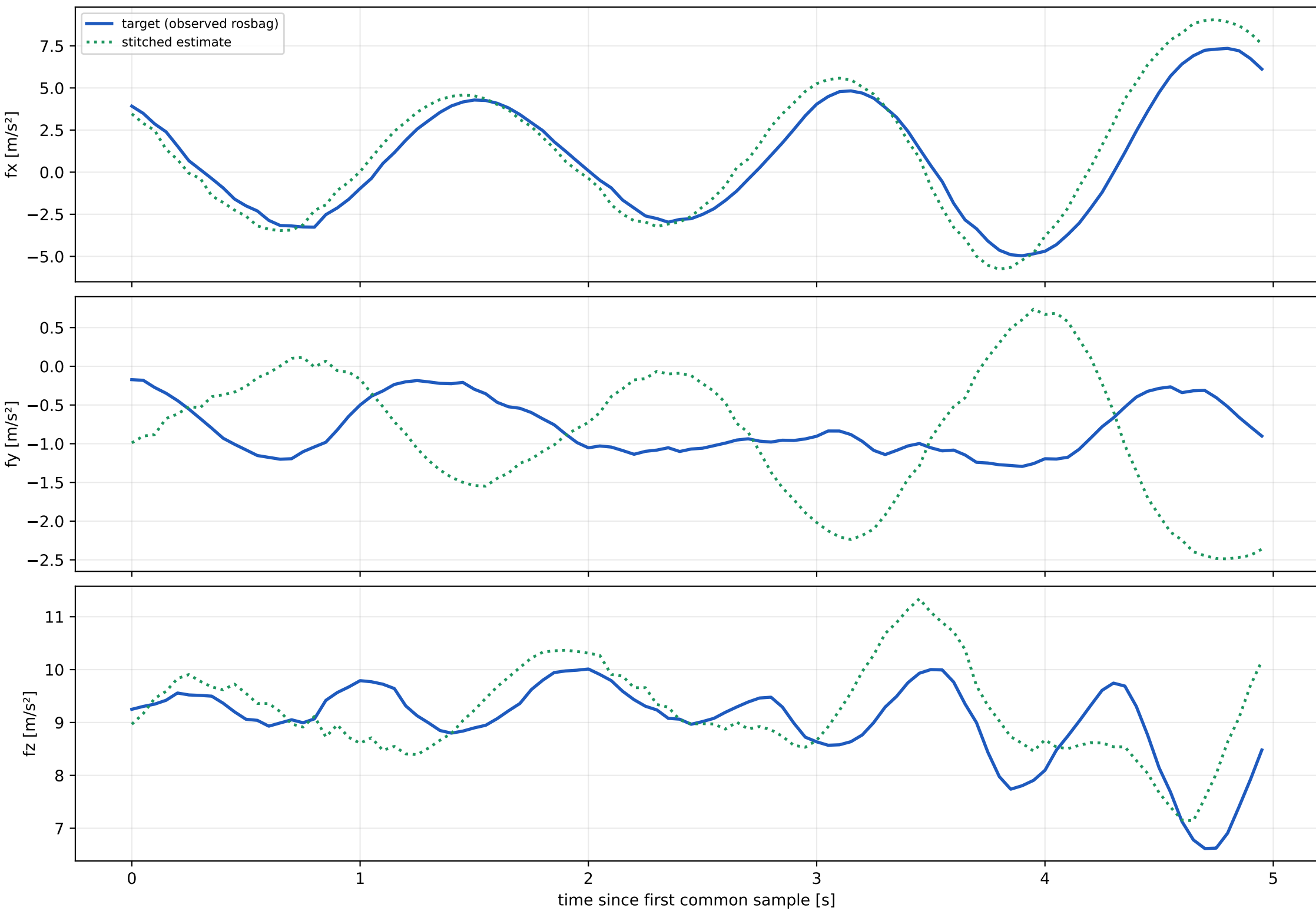
World-frame sensor velocity (diagnostic; excluded from fit)



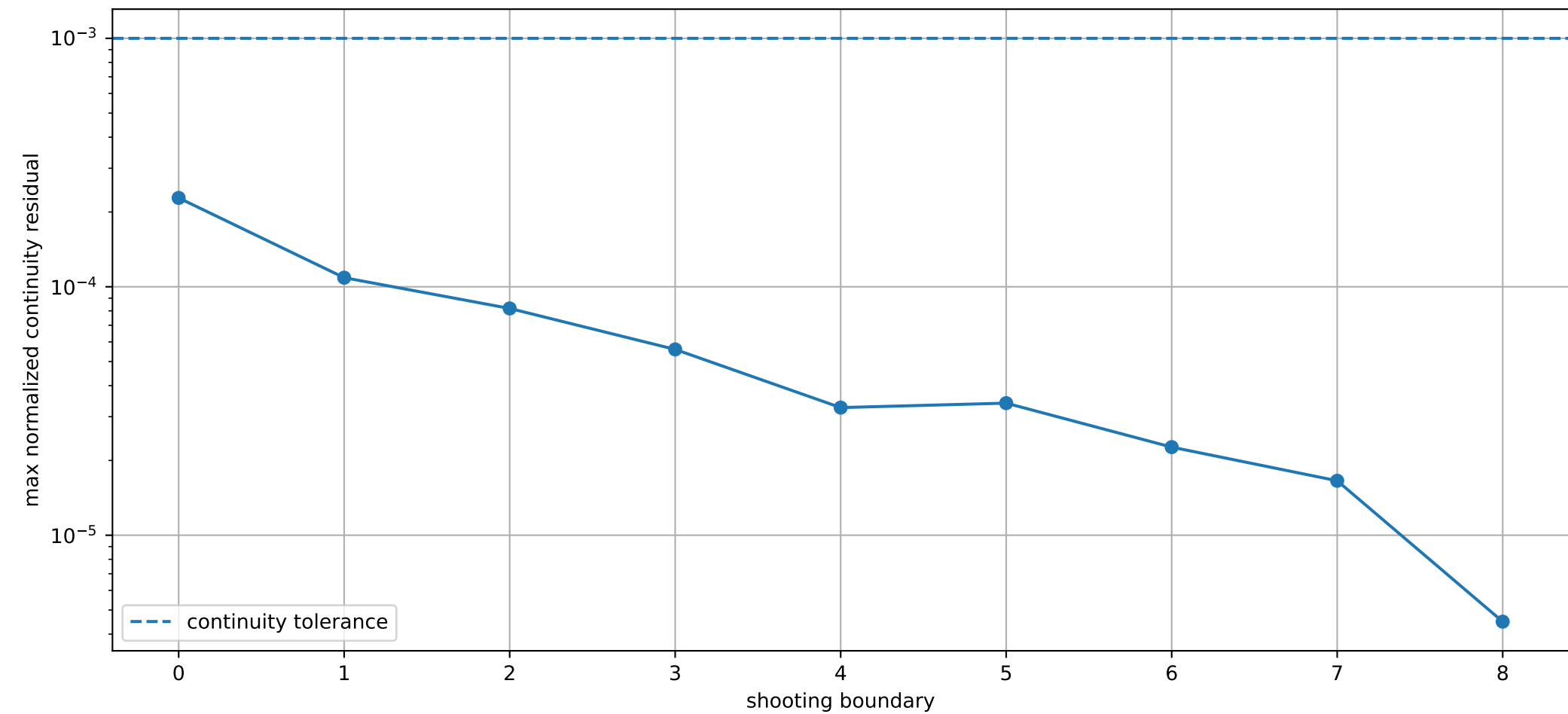
IMU angular velocity (diagnostic; excluded from fit)



IMU specific force (diagnostic; excluded from fit)



Shooting continuity diagnostic



Final estimated parameters

Selected multiple-shooting estimate

Decoded physical parameters

```
mass [kg] 9.75286282873
inertia [kg m^2]
[ 1.69660445774  0.627670673892 -0.268287300119]
[ 0.627670673892  0.749468337629  0.414967037786]
[-0.268287300119  0.414967037786  2.16668477699]
principal inertia [kg m^2] [0.30696605495, 1.99986581258, 2.30592570483]
CoG offset [m] [0.210101292691, 0.0858686202159, -0.0809356578849]
rotor force effectiveness [0.158739695831, 4.78509103525, 9.8327100037, 0.133890887703]
command delay [s] 0.02
```

Thirteen optimized smooth coordinates

```
log_mass_scale 1.42246594231
log_second_moment_cholesky_xx_scale 1.1234262214
log_second_moment_cholesky_yy_scale 1.32368590068
log_second_moment_cholesky_zz_scale 0.0200508152386
normalized_second_moment_cholesky_yx_offset -3.16506225339
normalized_second_moment_cholesky_zx_offset 4.60669152228
normalized_second_moment_cholesky_zy_offset -1.94935085346
cog_offset_x_m 0.210101292691
cog_offset_y_m 0.0858686202159
cog_offset_z_m -0.0809356578849
force_effectiveness_contrast_1 -2.40840187928
force_effectiveness_contrast_2 -1.97854009486
force_effectiveness_contrast_3 2.32179110776
```

Fit diagnostics

```
stitched SE(3) loss 23.8554073615
full-rollout SE(3) loss 23.8626774838
broad soft-prior cost 9.32082657744
position RMSE [m] 0.291098605355
orientation RMSE [deg] 20.9900498623
velocity RMSE [m/s] 0.84702613936
angular velocity RMSE 0.632211594953 rad/s
specific force RMSE 1.7998365618 m/s^2
continuity max (normalized) 0.000227851621283
continuity tolerance 0.001
continuity converged True
```